

Statistical Properties of Forecast-Selected Penalized Trend Estimators

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Abstract

Finite-difference penalized least-squares trend estimators are linear smoothers when their smoothing parameter is fixed. In that regime, their expectation, bias, covariance, effective degrees of freedom, influence matrix, and forecast variance follow directly from the smoothing matrix. The statistical object changes once smoothness is selected from the same time series by chronological forecast validation. The resulting estimator,

$$\hat{\tau}_{\hat{S}} = H_{\lambda(\hat{S})}y,$$

is no longer a fixed linear smoother because the selected smoothness $\hat{S}$ is itself random and data-dependent. This paper develops the fixed-smoothness foundation needed to study that distinction and formulates the main inferential questions created by forecast-driven smoothness selection: selection-adjusted complexity, bias and variance, uncertainty for trend functionals, uncertainty in forecast-optimal smoothness, and propagation of selection uncertainty into trend forecasts. The purpose is not to rederive classical linear-smoother theory as a novelty claim, but to identify precisely what is known conditionally on a fixed smoother and what must still be established for the forecast-selected estimator used in the companion smoothness-selection paper.

1 Motivation and separation from the companion papers

The research program contains three distinct questions.

The companion smoothness-selection paper asks

$$\hat{S} \in \arg \min_{S \in [0,1]} F(S),$$

where $F(S)$ is a chronological rolling future-block forecast loss. Its question is: *which amount of smoothness should be selected for forecasting?*

The numerical-methods paper assumes that objective is already defined and asks: *how can its relevant minima and global optimum be located reliably and efficiently when the objective is multimodal?*

The present paper asks a third question:

Once smoothness has been selected from the data, what statistical properties does the resulting trend estimator have, and how should its uncertainty be quantified?

For fixed S , penalized least squares is a linear smoother. After forecast-based selection,

$$y \mapsto \hat{S}(y) \mapsto H_{\lambda(\hat{S}(y))} \mapsto H_{\lambda(\hat{S}(y))}y$$

is a data-adaptive nonlinear mapping.

The central statistical boundary of this paper is therefore

$$\boxed{\text{fixed-smoother statistics} \quad \text{versus} \quad \text{statistics after forecast-driven smoothness selection.}}$$

2 Model and notation

Let

$$y = (y_1, \dots, y_N)^\top \in \mathbb{R}^N$$

be an observed time series. For difference order d , let

$$D_d \in \mathbb{R}^{(N-d) \times N}, \quad Q = D_d^\top D_d.$$

The baseline penalized least-squares estimator is

$$\hat{\tau}_\lambda = \arg \min_{\tau \in \mathbb{R}^N} \left\{ \|y - \tau\|_2^2 + \lambda \|D_d \tau\|_2^2 \right\},$$

with solution

$$\boxed{\hat{\tau}_\lambda = H_\lambda y, \quad H_\lambda = (I + \lambda Q)^{-1}.}$$

The baseline model is deliberately the zero-drift, identity-weight case used by the companion methodological papers. The broader Guerrero formulation with nonzero difference drift or a non-identity observation-weight/covariance structure is treated as an extension rather than silently mixed into the baseline.

Because

$$x^\top Q x = x^\top D_d^\top D_d x = \|D_d x\|_2^2 \geq 0,$$

Q is symmetric positive semidefinite. Under the standard finite-difference operator, its nullity is d . Thus, if

$$Q = U \operatorname{diag}(\underbrace{0, \dots, 0}_d, \delta_1, \dots, \delta_{N-d}) U^\top, \quad \delta_j > 0,$$

then

$$H_\lambda = U \operatorname{diag} \left(\underbrace{1, \dots, 1}_d, \frac{1}{1 + \lambda \delta_1}, \dots, \frac{1}{1 + \lambda \delta_{N-d}} \right) U^\top.$$

The normalized smoothness coordinate is

$$\boxed{S(\lambda) = 1 - \frac{1}{N-d} \sum_{j=1}^{N-d} \frac{1}{1 + \lambda \delta_j}.$$

It maps $\lambda = 0$ to $S = 0$, and the limiting $\lambda \rightarrow \infty$ model to $S = 1$.

3 What is already known for fixed smoothness

This section collects the statistical foundation that follows once λ , equivalently S , is treated as fixed. These identities are important for the present paper, but they should not be presented as new without a literature-specific audit because closely related results are standard in Whittaker–Henderson smoothing, smoothing splines, generalized ridge/Tikhonov regularization, Hodrick–Prescott filtering, Gaussian random-walk/GMRF models, and related state-space formulations.

3.1 Linear-smoother structure

For fixed λ ,

$$\hat{\tau}_\lambda = H_\lambda y$$

is linear in y . The matrix H_λ is symmetric positive semidefinite, and every eigenvalue lies in $(0, 1]$.

The penalized spectral directions are attenuated by

$$\alpha_j(\lambda) = \frac{1}{1 + \lambda \delta_j}.$$

The null-space directions are not attenuated.

Proposition 1 (Null-space reproduction). *If $x \in \ker(D_d)$, then*

$$H_\lambda x = x$$

for every $\lambda \geq 0$.

For the usual difference operator, $\ker(D_d)$ contains discrete polynomial sequences of degree at most $d - 1$.

3.2 Effective degrees of freedom and normalized smoothness

For a fixed linear smoother, define

$$\text{edf}(\lambda) = \text{tr}(H_\lambda).$$

Using the spectral representation,

$$\text{tr}(H_\lambda) = d + \sum_{j=1}^{N-d} \frac{1}{1 + \lambda \delta_j}.$$

Therefore

$$\boxed{\text{edf}(\lambda) = N - (N - d)S(\lambda).}$$

Equivalently,

$$\boxed{S = \frac{N - \text{edf}}{N - d}.$$

This identity gives a direct statistical interpretation of normalized smoothness: S is the normalized fraction of penalizable effective flexibility removed by the smoother. At the endpoints,

$$S = 0 \iff \text{edf} = N, \quad S = 1 \iff \text{edf} = d.$$

A central later question is whether $\text{tr}(H_{\hat{\lambda}})$ remains an adequate complexity measure after $\hat{\lambda}$ is selected from the data.

3.3 Expectation, bias, variance, covariance, and risk

Assume first

$$y = \tau + \varepsilon, \quad \mathbb{E}(\varepsilon) = 0, \quad \text{Var}(\varepsilon) = \sigma^2 I.$$

For fixed λ ,

$$\mathbb{E}(\hat{\tau}_\lambda) = H_\lambda \tau,$$

and

$$\boxed{\text{Bias}(\hat{\tau}_\lambda) = (H_\lambda - I)\tau.}$$

Likewise,

$$\boxed{\text{Var}(\hat{\tau}_\lambda) = \sigma^2 H_\lambda^2.}$$

More generally, if

$$\text{Var}(\varepsilon) = \Sigma,$$

then

$$\boxed{\text{Var}(\hat{\tau}_\lambda) = H_\lambda \Sigma H_\lambda^\top.}$$

For squared Euclidean risk,

$$\mathbb{E}\|\hat{\tau}_\lambda - \tau\|_2^2 = \|(H_\lambda - I)\tau\|_2^2 + \sigma^2 \text{tr}(H_\lambda^2).$$

3.4 Linear functionals of the trend

For any $a \in \mathbb{R}^N$,

$$a^\top \hat{\tau}_\lambda = a^\top H_\lambda y,$$

and

$$\text{Var}(a^\top \hat{\tau}_\lambda) = a^\top H_\lambda \Sigma H_\lambda^\top a.$$

This includes level, local slope, and local curvature. The difficult question is whether intervals constructed conditionally on a selected $\hat{S}$ have correct post-selection coverage.

3.5 Influence and residuals

For fixed λ ,

$$\frac{\partial \hat{\tau}_i}{\partial y_j} = (H_\lambda)_{ij}.$$

Hence H_λ is an observation-to-fit influence map.

Residuals are

$$e_\lambda = (I - H_\lambda)y,$$

with

$$\text{Var}(e_\lambda) = (I - H_\lambda)\Sigma(I - H_\lambda)^\top.$$

Unlike an OLS hat matrix, H_λ is generally not idempotent:

$$H_\lambda^2 \neq H_\lambda.$$

Thus OLS residual degrees-of-freedom formulas should not be copied mechanically.

4 Forecasting with fixed smoothness

Let $G_{d,h}$ denote the fixed continuation matrix. Then

$$\hat{z}_\lambda = G_{d,h} H_\lambda y.$$

Conditional on fixed λ ,

$$\mathbb{E}(\hat{z}_\lambda) = G_{d,h} H_\lambda \tau,$$

and

$$\text{Var}(\hat{z}_\lambda) = G_{d,h} H_\lambda \Sigma H_\lambda^\top G_{d,h}^\top.$$

This is uncertainty in the estimated and extrapolated trend. A full predictive distribution additionally requires future innovation or observation noise.

After smoothness selection there is a third source:

forecast uncertainty = trend-estimation uncertainty + future innovation uncertainty + smoothness-selection uncertainty

with the caveat that exact finite-sample variance need not decompose additively without covariance terms.

5 The new statistical object: forecast-selected smoothness

The smoothness-CV paper selects

$$\hat{S} \in \arg \min_{S \in [0,1]} F(S).$$

The resulting estimator is

$$\hat{\tau}_{\hat{S}} = H_{\lambda(\hat{S})} y.$$

The notation is deceptively similar to the fixed- S estimator, but

$$\hat{S} = \hat{S}(y)$$

is random. Consequently,

$$y \mapsto H_{\lambda(\hat{S}(y))} y$$

is generally nonlinear.

This is the central source of nontriviality in the paper.

5.1 Why fixed-smoother variance is not the whole answer

Conditioning on $\hat{S} = s$, one can write

$$H_{\lambda(s)} \Sigma H_{\lambda(s)}^\top.$$

But if inference concerns the full procedure that first selects S and then estimates the trend, the randomness of $\hat{S}$ must also be accounted for.

The same warning applies to effective degrees of freedom, confidence bands, tests of slope or curvature, and forecast uncertainty.

6 What we want to know

The following are research targets, not established novelty claims.

6.1 Sampling behavior of selected smoothness

We want to characterize

$$\hat{S} = \arg \min_S F(S)$$

as a random quantity:

- its sampling or stability distribution;
- dependence on horizon, sample length, residual dependence, noise scale, and latent roughness;
- uncertainty when $F(S)$ is multimodal;
- confidence sets for a clearly defined population/oracle forecast-optimal S^* .

6.2 Selection-adjusted effective degrees of freedom

For fixed λ ,

$$\text{edf} = \text{tr}(H_\lambda).$$

For

$$m(y) = H_{\lambda(\hat{S}(y))}y,$$

we want an effective complexity measure for the full adaptive mapping rather than assuming

$$\text{edf}_{\text{adaptive}} = \text{tr}(H_{\hat{\lambda}}).$$

6.3 Bias and variance after selection

We want to understand

$$\mathbb{E} \left[H_{\lambda(\hat{S})} y \right]$$

and

$$\text{Var} \left(H_{\lambda(\hat{S})} y \right),$$

not only their conditional fixed- S analogues.

Candidate approaches include exact results in simplified Gaussian settings, local expansions around stable interior optima, bootstrap approximations, and simulation-based coverage studies.

6.4 Inference for level, slope, and curvature

We want selection-aware inference for quantities such as

$$e_t^\top H_{\lambda(\hat{S})} y, \quad \Delta H_{\lambda(\hat{S})} y, \quad \Delta^2 H_{\lambda(\hat{S})} y.$$

Scientifically useful targets include level, local slope, sign of slope, curvature, turning-point location, and terminal slope/curvature used by extrapolation.

6.5 Propagation of smoothness uncertainty into forecasts

For fixed λ ,

$$\hat{z}_\lambda = GH_\lambda y.$$

Since

$$\frac{\partial H_\lambda}{\partial \lambda} = -H_\lambda Q H_\lambda,$$

we have

$$\frac{\partial \hat{z}_\lambda}{\partial \lambda} = -GH_\lambda Q H_\lambda y.$$

Because S is monotone in λ ,

$$\boxed{\frac{\partial \hat{z}}{\partial S} = -\frac{GH_\lambda Q H_\lambda y}{S'(\lambda)}}.$$

This suggests a local delta-method term involving

$$\left(\frac{\partial \hat{z}}{\partial S}\right) \text{Var}(\hat{S}) \left(\frac{\partial \hat{z}}{\partial S}\right)^\top,$$

but its validity must be investigated. It may be poor near boundaries, flat objectives, or switches between local minima.

6.6 Multimodality as a statistical issue

The numerical-methods paper treats multiple minima as an optimization problem. Here the same phenomenon is statistical.

If two minima S_1 and S_2 have similar forecast loss but imply substantially different effective degrees of freedom, terminal derivatives, or forecast geometry, small perturbations may move the fitted procedure between different smoothing regimes.

We want to characterize competing minima by

edf, roughness, bias/variance, spectral attenuation, terminal derivatives, forecast geometry.

7 A local expansion around a stable selected optimum

Suppose the relevant optimum is unique, interior, and locally stable:

$$\hat{S} = S^* + \Delta_S.$$

Then formally,

$$\hat{\tau}_{\hat{S}} \approx \hat{\tau}_{S^*} + \left. \frac{\partial \hat{\tau}_S}{\partial S} \right|_{S^*} \Delta_S.$$

Because

$$\frac{\partial \hat{\tau}}{\partial \lambda} = -H_\lambda Q H_\lambda y,$$

$$\boxed{\frac{\partial \hat{\tau}}{\partial S} = -\frac{H_\lambda Q H_\lambda y}{S'(\lambda)}}.$$

This suggests a first-order decomposition into fixed- S^* variability and smoothness-selection variability. However, $\hat{S}$ and y are generated from overlapping data, so covariance terms matter. A rigorous treatment must derive rather than assume the decomposition.

The expansion is not expected to describe boundary solutions or local-minimum switching well.

8 What would constitute a contribution

The following are foundations, not sufficient contributions by themselves:

- $\text{edf} = \text{tr}(H)$;
- fixed- λ bias and variance;
- leverage from H_{ii} ;
- ordinary fixed- λ Gaussian intervals;
- the Bayesian interpretation of a quadratic penalty;
- relationships with HP, ridge/Tikhonov, or Whittaker smoothing;
- polynomial continuation implied by the null space.

A publishable contribution should center on one or more nontrivial consequences of forecast-driven selection:

1. selection-adjusted degrees of freedom;
2. uncertainty and coverage after forecast-CV smoothness selection;
3. sampling/stability theory for $\hat{S}$;
4. treatment of multimodal smoothness-selection uncertainty;
5. forecast-risk decomposition including selection uncertainty;
6. inference for terminal level/slope/curvature after selection;
7. horizon-dependent extrapolation-risk theory.

The final paper should choose one or two primary theorem and simulation targets rather than claiming all of these at once.

9 Relationship to the smoothness-CV paper

The distinction is

smoothness-CV: $y \mapsto \hat{S}$

versus

statistical-properties: $(y, \hat{S}) \mapsto \hat{\tau}_{\hat{S}} \mapsto \text{bias, variance, complexity, and uncertainty.}$

The smoothness-CV paper may use simple fixed-smoother properties to interpret S , especially

$$\text{tr}(H) = N - (N - d)S.$$

It should not absorb the full post-selection inference problem.

10 Relationship to the numerical-methods paper

The numerical paper answers how to compute candidate minima of $F(S)$. The present paper treats the selected minimum as a random output of a data-dependent procedure.

Thus:

criterion $\longrightarrow$ numerical solution $\longrightarrow$ statistical consequences.

Multimodality is the main bridge: numerically it means several minima must be found; statistically it means the selected estimator may jump between substantially different smoothing regimes.

11 Planned simulation program

The first simulations should be designed for inference rather than forecasting performance. The latent trend should be known so that bias, variance, coverage, and selection behavior can be measured.

Controlled factors should include:

- sample length;
- forecast horizon;
- observation-noise scale;
- latent-trend roughness;
- residual persistence;
- distance of the optimum from $S = 0$ or $S = 1$;
- unimodal versus multimodal forecast-CV geometry.

At minimum, compare:

1. an oracle fixed- S estimator;
2. a fixed but misspecified S ;
3. forecast-CV-selected $\hat{S}$;
4. conditional intervals that ignore selection;
5. one or more selection-aware uncertainty procedures.

Primary endpoints should include bias, variance, mean-squared error, coverage, interval width, effective-complexity estimates, the distribution/stability of $\hat{S}$, and forecast coverage where relevant.

12 Literature boundary

A dedicated literature audit is required before any novelty claim. The closest bodies of theory include Whittaker–Henderson smoothing, smoothing splines and P-splines, generalized ridge/Tikhonov regularization, Hodrick–Prescott filtering, Gaussian random walks and GMRFs, state-space trend models, generalized degrees of freedom, data-adaptive smoothing, selective inference, and uncertainty after tuning-parameter selection.

The repository already contains especially relevant material on Guerrero’s finite-difference PLS/controlled-smoothness formulation and modern Whittaker–Henderson treatments. These make clear that fixed-smoother effective degrees of freedom, smoothing bias, Bayesian uncertainty, and extrapolation uncertainty cannot by themselves be claimed as new here.

The novelty audit should therefore be organized around the narrower question:

What statistical theory already exists for a smoother whose tuning parameter is selected specifically by chronological forecast loss, especially when the validation surface is dependent, horizon-specific, and potentially multimodal?

13 Current status

Established within the repository:

- the finite-difference PLS estimator $H_\lambda y$;
- normalized smoothness $S(\lambda)$;
- $\text{edf} = N - (N - d)S$ for fixed smoothness;
- first and second derivatives of H_λ ;
- first and second derivatives of forecast loss;
- the forecast operator GH_λ ;
- empirical evidence that forecast-CV surfaces can be multimodal;
- a companion numerical method for recovering relevant minima.

Not yet established:

- the distribution and uncertainty of $\hat{S}$;
- selection-adjusted effective degrees of freedom;
- bias and variance of the full selected procedure;
- valid post-selection intervals for level/slope/curvature;
- a rigorous decomposition of forecast uncertainty including smoothness selection;
- treatment of multimodal smoothness-selection uncertainty.

These open items, rather than the fixed-smoother formulas alone, define the scientific target.

14 Conclusion

Finite-difference penalized trend estimation has a simple statistical description when smoothness is fixed. That simplicity provides an exact baseline against which the forecast-selected estimator can be measured. The main statistical challenge begins when smoothness is selected from chronological forecast performance. Then the fitted trend is no longer a fixed linear smoother, and its complexity, bias, variance, interval coverage, and forecast uncertainty must reflect the randomness and possible multimodality of the selection step.

The intended contribution is therefore not a new derivation of classical smoothing identities. It is a statistical theory and inferential framework for the forecast-selected penalized trend estimator produced by the companion smoothness-CV procedure.